

SUJAL RAJBHANDARI

AI/ML Engineer | 3+ Years Experience

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SUMMARY

AI/ML Engineer at **Resimator Oy** leading 7+ **production AI systems** across Generative AI, RAG, LangGraph multi-agent workflows, MCP server architectures, LLM-powered document extraction, recommendation systems, forecasting, and anomaly detection. Deployed end-to-end on **AWS** (EC2, S3, Lambda, ALB, Redis) using FastAPI and Docker.

EDUCATION

Presidential Graduate School, B.Sc. in Information Technology (BSIT)

Concentration in Artificial Intelligence & Machine Learning Kathmandu, Nepal

- **Grade Point Average (GPA): 3.92 / 4.0**
- **Concentration:** Artificial Intelligence and Machine Learning.

PROFESSIONAL EXPERIENCE

Resimator Oy July. 2024 – Jun. 2026

AI/ML Engineer (Hybrid) Remote

- Architected AI-powered document extraction systems for invoices and legal lease agreements using LLMs (Gemini, OpenAI), OCR (Tesseract, PyMuPDF), and Pydantic-validated structured JSON outputs, deployed via FastAPI on AWS.
- Engineered multi-agent AI workflows using LangGraph for stateful, graph-based orchestration enabling tool usage, reasoning, and multi-step task automation.
- Designed an **MCP server architecture** enabling AI agents and chatbots to dynamically access external tools, APIs, and business data for context-aware responses.
- Built an AI Business Analytics chatbot on top of the MCP server delivering KPIs, trend detection, AI insights, and chart-ready outputs from natural-language queries.
- Developed RAG pipelines using LangChain, vector databases (Pinecone, Chroma, FAISS, Qdrant), and embedding models for document-grounded QA with reduced hallucination.
- Built recommendation systems combining collaborative filtering, content-based filtering, hybrid scoring, and embedding-based similarity, evaluated with Precision@K, Recall@K, and NDCG@K.
- Developed anomaly detection, forecasting, image-generation, and web-scraping NLP pipelines for time-series prediction, irregular-pattern detection with LLM reasoning, and large-scale structured data extraction.
- Deployed production AI systems on AWS using EC2, S3, Lambda, ALB, Redis, Docker, and REST APIs, managing the full model serving and deployment lifecycle.

Creatu Developers Dec. 2022 – Mar. 2023

Data Science Intern Kathmandu, Nepal

- Performed EDA, data cleaning, and feature preparation on real-world datasets using Pandas, NumPy, Matplotlib, Seaborn, and Plotly.
- Built time-series prediction models for business trend analysis and short-term forecasting; developed interactive dashboards for stakeholder reporting.

SELECTED PROJECTS

RAG Chatbot with LangGraph & Vector DBs | *LangGraph, LangChain, Pinecone, Chroma, OpenAI, Gemini, FastAPI.*

- Retrieval-augmented chatbot with multi-agent orchestration and vector retrieval for grounded, context-aware responses.

MCP-Based AI Business Analytics Chatbot | *MCP, LangGraph, OpenAI, FastAPI, Qdrant.*

- Chatbot that dynamically fetches business data via MCP, answering analytical queries with KPIs, trends, and chart-ready outputs.

AI-Powered Invoice Extraction System | *Gemini, OpenAI, Tesseract OCR, Pydantic, FastAPI, AWS.*

- End-to-end LLM-based extraction from unstructured PDFs/images with Pydantic-validated JSON outputs exposed via REST APIs.

Legal Lease Agreement Information Extraction | *Gemini, NLP, OCR, pdf2image, PyMuPDF, FastAPI.*

- Extracts clauses, parties, dates, rent, deposits, and obligations from scanned and image-based legal PDFs into structured outputs.

Hybrid Recommendation System | TensorFlow, FastFM, FAISS, Two Tower, NCF, Three Stage:

- Built a restaurant recommendation system combining collaborative filtering, content-based retrieval, and popularity ranking to generate personalized menu recommendations and improve item discovery.

Anomaly Detection & Reasoning System | *Python, Scikit-learn, Pandas, OpenAI, FAISS*

- Designed an anomaly detection pipeline that combines statistical and ML-based detection methods with LLM-generated reasoning, allowing users to understand why unusual patterns were flagged.

TECHNICAL SKILLS

Languages: Python, SQL

Generative AI & LLMs: LangChain, LangGraph, RAG, MCP, Multi-Agent Systems, OpenAI API, Gemini API, Prompt Engineering, Structured JSON Extraction

ML / DL: Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, Recommendation Systems, Forecasting, Anomaly Detection

Vector DBs & Doc AI: Pinecone, Chroma, FAISS, Qdrant, Tesseract OCR, PyMuPDF, pdf2image, Pydantic

Cloud & MLOps: AWS (EC2, S3, Lambda, ALB), Redis, Docker, FastAPI, REST APIs

Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, Plotly, EDA, MySQL, PostgreSQL

Tools: Git, GitHub, Bitbucket, Jupyter, VS Code, Postman, ClickUp

CERTIFICATIONS

- Data Science • Unsupervised Learning, Recommenders & Reinforcement Learning • MCP with AI Agents • Vector Databases with RAG • Developing Generative AI Applications.